JAVIER PEREZ TOBIA

APPLIED AI / MACHINE LEARNING ENGINEER | RESEARCH & PRODUCTION DEPLOYMENT

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PROFILE

Applied AI/ML engineer and researcher with experience owning AI systems from problem framing and technical discovery through modeling, optimization, productionization, and enterprise adoption. Deep Python/AI-stack and mathematical modeling background, with strong customer-facing technical leadership in ambiguous environments. At QuantumBlack / McKinsey, delivered production ML, digital-twin, optimization, and data-platform transformations across energy, banking, and telecom, with deployed models and new ways of working contributing to tangible annual profit uplift. Produced three publications during research at UBC spanning adversarial robustness, biomolecular ML and time-series trace reconstruction.

TECHNICAL EXPERTISE

ML & research stack: **Python, PyTorch, JAX, Qiskit**, scikit-learn, NumPy, pandas; with practical research experience on deep learning, computer vision, time series, generative modeling, adversarial robustness, and biomolecular ML

Generative AI & LLM systems: agentic co-development and harnessing, RAG, agentic workflows, embeddings and vector retrieval, tool/function calling, LLM evaluation, and production integration of LLM-enabled applications

Optimization & control systems: **Gurobi, MATLAB, C++**; with industry deployments on stochastic and multi-asset optimization, portfolio optimization, adaptive model predictive control, digital twins, and mathematical modeling

Engineering & production: **SQL, Linux, Git, AWS, Azure**; MLOps, model serving and productionization, data pipelines and data-lake architecture, real-time ML systems, and enterprise AI platform design.

PROFESSIONAL EXPERIENCE

Associate – Data Science **Sep 2023 – Present**
QuantumBlack, AI by McKinsey *Madrid, Spain*

Data Scientist (Sep 23–Nov 24) → Sr. DS (Nov 24–Nov 25) → Associate (DS) (Nov 25–Present).

Owned end-to-end AI and analytics workstreams and led DS teams of 5+ people across energy, banking, and telecom, translating ambiguous business problems into data/ML architectures, models, production systems, and operating-model changes across multiple geographies. Namely:

- **AI transformation of energy trading:** developed ML forecasting and Gurobi-based multi-asset optimization capabilities, end-to-end process orchestration, and deployment approaches that embedded AI into core trading decisions while building capabilities across software development teams.
- **Digital transformation of refining operations:** built digital twins, real-time ML models for crude-property prediction, and optimization modules; integrated analytics into operational workflows and new ways of working.
- **ML-driven retail-bank deposit pricing:** developed price-sensitivity and portfolio-allocation models; proofs of concept demonstrated measurable P&L impact in under four weeks, followed by wave-based deployment across the managed portfolio.
- **Customer-value optimization:** redesigned analytics for major Latin American and Iberian telecoms, combining ML targeting models and LLM based solutions for call transcription analytics and knowledge retrieval, data-lake architecture redesign, and a company-wide optimization engine for automated segmentation, targeting, and campaign execution.

RESEARCH EXPERIENCE

Machine Learning Researcher **May 2021 – Jul 2022**
University of British Columbia – DAIS & IDSL Labs *Kelowna, Canada*

- **Robust deep learning / computer vision:** developed architectures and attribution-guided defenses to improve neural-network robustness against adversarial examples; achieved state-of-the-art results against AutoAttack on benchmark datasets. **IDA 2022**
- **Bioinformatics / biomolecular ML:** built models for aptamer–protein binding and methods for aptamer sequence search and generation with experimentally validated predicted sequences. **ACS Synthetic Biology 2023**
- **Time-series / trace reconstruction and imbalanced learning:** conducted research on event-trace reconstruction and data-generation methods for imbalanced classification. **IEEE SMC 2021**

EDUCATION & HONORS

B.Sc. – Physics & Mathematics **2018 – 2023**
University of British Columbia *Kelowna, Canada*

- **Academic performance:** 96.2% cumulative average.
- **Selected honors:** Graduating Student Award in Physics & Mathematics; International Student Faculty Award; International Major Entrance Scholarship; Outstanding International Student.
- **QHack 2022 – Quantum Computing Hackathon:** ranked **#29 globally** out of +800 participants; completed all 25 coding challenges spanning quantum algorithms, quantum machine learning, and quantum chemistry using Qiskit and PennyLane.
- **CAN-SBX Winner – Canadian Stratospheric Balloon Experiment:** co-led the winning undergraduate team designing a stratospheric particle-flux experiment subsequently flown on Canadian Space Agency campaigns.
- **Academic leadership:** Co-founder, Student Independent Research Club; Teaching Assistant for undergraduate mathematics courses.

LANGUAGES

Spanish: native | English: C2 / bilingual proficiency | French: B2 / intermediate proficiency